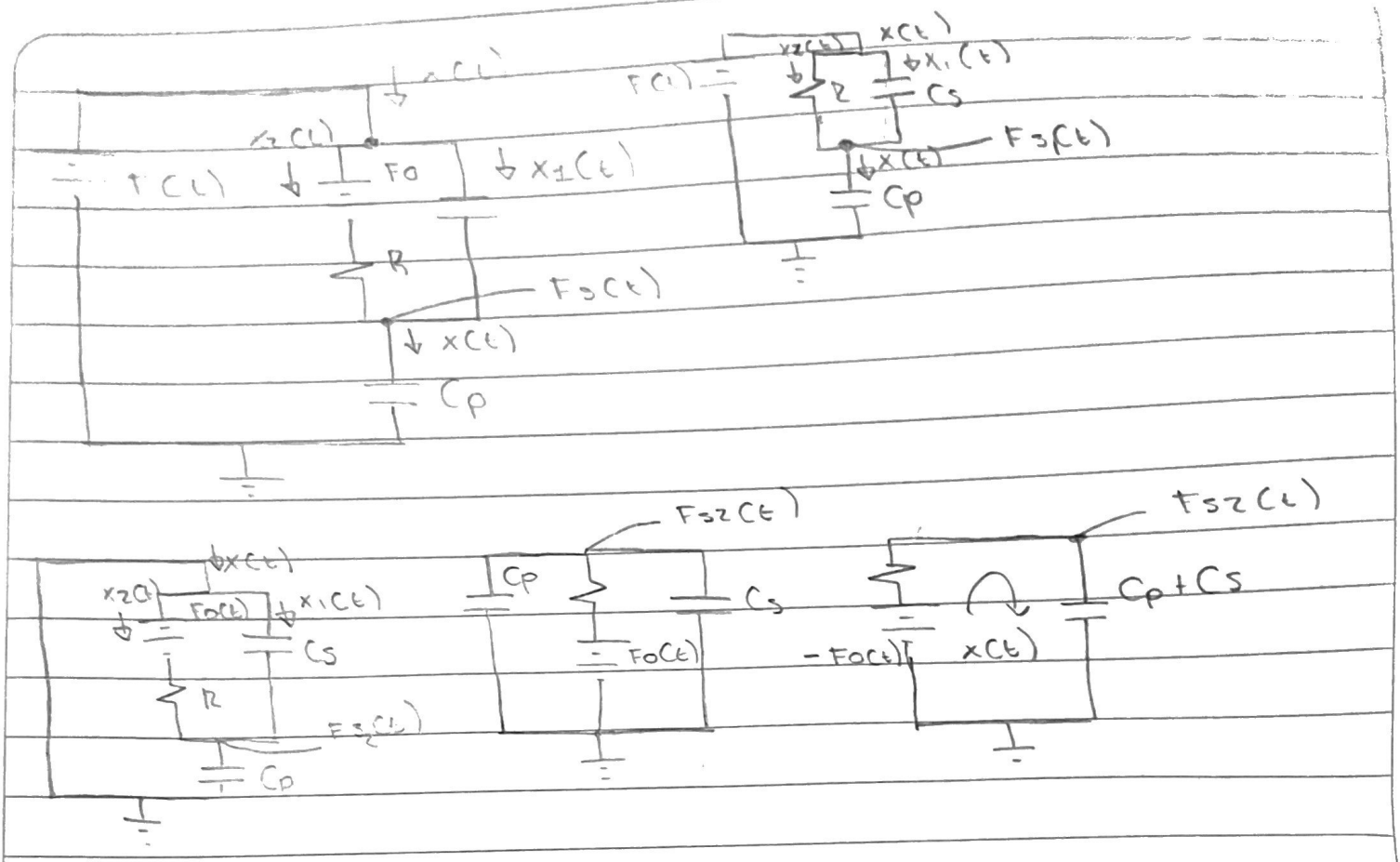


Circuito 1. Apagando $F_o(t)$

$$x(t) = X_1(t) + X_2(t)$$

$$x(t) = C_p \frac{dF_{s1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) + C_s s F_{s1}(s) + \frac{F_{s1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} \cdot F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2. Apagando $F(t)$

Recordando que $F_o(t) = a\,F(t)$

$$-aF(t) = R\,x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$-a\,f(s) = R\,x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$f(s) = -\frac{1}{a}\left(R + \frac{1}{(C_p + C_s)s}\right)x(s) = -\frac{1}{a}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\,x(s)$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{a}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\,x(s)} = -\frac{a}{R(C_p + C_s)s + 1}$$

$$F_{s2}(s) = -\frac{a\,F(t)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s}\,F(s) - \frac{a\,F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - a}{R(C_p + C_s)s + 1}\,F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - a}{R(C_p + C_s)s + 1}$$

Errores en estado estacionario

$$e(s) = \lim_{s \to 0} s\,F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} s\,\frac{1}{s}\left(1 - \frac{R C_s s + 1 - a}{(R C_p + R C_s)s + 1}\right) = a$$

Estabilidad en lazo abierto

$(R C_p + R C_s)s + 1 = 0$ El sistema es estable

$$\lambda = \frac{1}{R C_p + R C_s}$$